Retrieve-then-Verify: an LLM Verifier for Streaming Linking

Step 8 — Task 2b, CLAUDE.md Milestone 3b

Tomer Morad Or Israeli Yoel Marko

July 2026

1. Objective

Step 7 showed that no fixed cosine threshold on e5_k1 similarity can separate rare true repeats from the flood of similar-sounding unrelated topics at gold-segmentation granularity (best-F1 streaming precision ≈8%). This step tests the project brief's proposed fix: use similarity only to *retrieve* a candidate shortlist, then have an **LLM verifier** read the actual text and decide same/related/new, rather than trusting a vector-distance cutoff. **Status: paused below baseline, not dead-ended** — every configuration tried is documented here, including several that made things worse, because the negative results are as informative as the positive ones for scoping future work.

2. Method

Retrieval side. An oracle-growth journal (new entry created on each topic's true first occurrence) ranks existing entries by e5_k1 cosine similarity for all 36 gold repeat events.

Verifier side. 522 streaming-decision queries, each with the new segment's text and its top-10 retrieved candidates (100-word content-only snippets, deliberately excluding the gold topic label to prevent string-matching the answer). Three local HF chat models compared: `dicta-il/dictalm2.0-instruct`, `Qwen2.5-7B-Instruct`, `Qwen2.5-3B-Instruct`. The model picks the best-matching candidate rank and a 0–100 confidence score, swept as a threshold exactly like Step 7's cosine curve for direct comparability.

3. Results

3.1 Retrieval recall@K — never the bottleneck

K	1	2	3	5	10
Recall	0.694	0.778	0.917	0.944	**1.000**

All 36/36 gold repeats are retrievable in the top 10. Three embedding/lexical fixes for the 69% recall@1 gap (regex boilerplate stripping, TF-IDF rerank, ministry-name gating) were each tried and ruled out with evidence (ceiling ≈+1/36 in all three cases) — the corpus's recurring "year-end budget surplus" sessions share so much administrative vocabulary across *different* ministries that lexical tricks cannot cleanly separate them; one case needs genuine reading comprehension (same ministry, same section, different specific matter). This motivated committing to an LLM verifier rather than continuing to chase retrieval fixes.

3.2 Verifier comparison — 6 real bugs fixed en route

Getting to a working, comparable verifier required: (1) merging system+user roles for DictaLM's chat template, (2) stopping models from literally continuing the (likely memorized) real transcript text past the prompt cutoff, (3–4) anchoring the exact JSON schema with a worked example without `no_repeat_ngram_size` banning the model from legitimately repeating that schema's tokens, (5) discovering a fixed same/new decision was miscalibrated completely differently per model, fixed by (6) switching to a swept 0–100 confidence score.

Model	Best F1	Precision	Recall	θ
dictalm2	0.086	4.5%	92.0%	0
qwen3b	**0.111**	6.0%	70.4%	65
qwen7b	0.077	4.0%	95.0%	15
Baseline (e5_k1 threshold, Step 7)	**0.125**	7.7%	33.3%	—

Recall is excellent across all three models (70–95%, vs. baseline's 33%) — they genuinely find almost every true recurrence. **Precision is the isolated bottleneck.**

3.3 Four precision-calibration experiments

Experiment	qwen3b F1	qwen7b F1	dictalm2 F1
baseline	**0.111**	0.077	0.086
Ensemble (majority vote, no new LLM calls)	0.109 — ties qwen3b, doesn't beat it		
+more few-shot examples (2→4)	0.104 ↓	0.070 ↓	0.057 ↓
+two-score schema (topical vs. specific)	0.102 ↓	**0.085** ↑	**0.096** ↑
+longer snippets (100→400 words)	0.068 ↓↓	0.055 ↓↓	0.000 (broke)

None beat the baseline. Ensemble majority-voting successfully filters dictalm2/qwen7b's excess false positives (precision rises to 5.9% vs. their 4.0–4.5%) but cannot exceed the best individual model once it's one of the three votes. Longer snippets actively hurt every model and broke dictalm2 outright (its old document-continuation failure mode resurfaces at longer context). More few-shot examples also hurt across the board. The two-score schema is the one genuinely interesting result: it helped the two *more miscalibrated* models (dictalm2, qwen7b) but slightly hurt qwen3b, which was already well-calibrated.

4. What Needs Improvement

1. **The precision ceiling is real and evidenced, not a quick prompt-engineering fix.** Four different calibration ideas were tried; none closed the gap to baseline.
2. **Two separate scores helped the weaker models** — worth exploring further with the two-score schema *and* extended few-shot together (not yet combined), or with a larger model.
3. **Whether the 100-word snippet cap is fundamentally too little context**, or whether longer context is inherently harmful for this classification-style task (unlike Step 9's free-text generation, where more context per candidate was never tried) remains genuinely open.
4. **"Related" predictions are unused.** The current schema only distinguishes LINK vs. NEW for scoring, since gold has no 3-way label. Whether "related" is usable for Task 3 (log writing) input is still open.
5. **Recommendation for now**: keep method (a), the similarity-threshold baseline, as the production linking method (decision made 2026-07-15) — it remains the best of everything tried, including this step's LLM verifier.

Artifacts (this step):

`outputs/shortlist_recall.json`, `outputs/verifier_queries.json`,
`outputs/verifier_predictions_*.json` (14 configs), `outputs/verifier_eval_summary.json`
Code: `code/build_verifier_inputs.py`, `code/run_verifier.py`, `code/eval_verifier.py`,
`code/ensemble_verifier.py`,
`code/shortlist_recall.py`, `code/rerank_recall.py`, `code/ministry_gate_recall.py`,
`code/strip_procedural.py`, `code/inspect_misses.py`.